

EXEC-04: PRESS RELEASE — SERIES A FUNDING AND PUBLIC LAUNCH

Embargoed Until: [Date TBD — Coordinated with Series A Close] **Document Ref:** CSM-EXEC-04-v1.0 **Draft Date:** July 15, 2026

FOR IMMEDIATE RELEASE

CARRINGTON STORM MOTORS SECURES \$25 MILLION SERIES A TO PROTECT GLOBAL ELECTRICAL GRID FROM CATASTROPHIC SOLAR STORMS

Montana-Based Deep Technology Company Unveils Dielectric Citadel™ — A Complete Protection Architecture Built From 6,302 Research Documents, 50+ Engineered Products, and a 200-Country Diplomatic Framework — Backed by MXene Materials Science, Ceramic Composite Engineering, and a Groundbreaking Insurance Industry Partnership Model

BOZEMAN, Montana — [Date] — Carrington Storm Motors (CSM), a deep technology company dedicated to protecting civilization’s electrical infrastructure from the catastrophic effects of solar coronal mass ejections — known as Carrington Events — today announced the close of a \$25 million Series A funding round and its formal public launch. The funding will accelerate the transition from CSM’s fully developed research and design phase into initial production, regulatory certification, and the deployment of pilot installations at five U.S. electric utility partner sites.

The company also unveiled the Dielectric Citadel™, a comprehensive protection architecture encompassing 50-plus engineered products, 12 volumes of materials science research, 72 volumes of fleet specifications, a 200-country diplomatic engagement framework, and 20 insurance industry dossiers — all developed over four years by CSM’s unique 18-agent cognitive delivery system.

“The electrical grid is the single most valuable physical asset humanity has ever built, and it is vulnerable to an event that has occurred before and will occur again,” said Kairos Steele, Director and Founder of Carrington Storm Motors. “A Carrington-scale solar storm — the kind that set telegraph stations on fire in 1859 — would destroy hundreds of high-voltage transformers within 90 seconds of impact, triggering a cascade of infrastructure failures that could leave large portions of the developed world without electricity for years. This is not a hypothetical scenario. It is a well-modeled, actuarially priced, geologically documented risk. We have built the technology to protect against it. This funding allows us to deploy it.”

THE THREAT

A Carrington Event occurs when a coronal mass ejection from the Sun strikes Earth’s magnetosphere, inducing geomagnetic currents in long-distance electrical conductors — primarily, the 5.5 million kilometers of high-voltage transmission lines in the United States alone. These geomagnetically induced currents (GIC) can saturate transformer cores, causing internal heating that destroys winding insulation in as little as 90 seconds.

The 1859 Carrington Event — the most powerful geomagnetic storm ever recorded — caused telegraph systems across North America and Europe to fail, spark, and in some cases catch fire. In 1989, a storm roughly one-tenth the magnitude of the Carrington Event collapsed the entire Hydro-Québec grid in 90 seconds, leaving six million people without power.

The economic exposure is staggering. The 2008 National Academy of Sciences report “Severe Space Weather Events” estimated recovery from a Carrington-scale event could take 4 to 10 years and cost \$1 to \$2 trillion. A 2013 Lloyd’s of London study placed the exposure at \$0.6 to \$2.6 trillion for the North American grid alone. Updated analyses incorporating cascading infrastructure failures place the total at \$4 to \$6 trillion for the United States alone, with global exposure exceeding \$10 trillion.

According to CSM’s analysis of 41 insurance carrier internal assessments, the global insurance industry has independently identified space weather as one of the largest unhedged catastrophic risks in financial history, yet no

comprehensive protection solution has been brought to market — until now.

THE SOLUTION: THE DIELECTRIC CITADEL

CSM's Dielectric Citadel is built on five core materials-science innovations:

MXene-Polymer Dielectric Composites: Two-dimensional transition metal carbide materials — MXenes — combined with proprietary polymer matrices to create flexible, lightweight shielding that blocks 65-72 decibels of electromagnetic interference across the full GIC frequency spectrum at just 0.5mm thickness. Unlike traditional Faraday cage approaches, MXene composites can be extruded, injection-molded, or applied as field-installable wraps to existing infrastructure.

Ceramic-Matrix Composite Transformer Housing: A silicon carbide fiber-reinforced aluminum oxide composite that replaces traditional steel transformer housings with a material that combines 450 MPa compressive strength with 10^{12} ohm-centimeter electrical resistivity — effectively an electrical insulator with the structural properties of steel. This housing prevents geomagnetic currents from reaching the transformer core entirely.

Geopolymer Concrete Substation Foundations: A fly-ash-based concrete alternative that achieves electrical resistivity five orders of magnitude higher than standard Portland cement when saturated, while reducing carbon footprint by 80 percent compared to conventional concrete. Substation foundations are a major GIC conduction pathway; replacing them with geopolymer concrete interrupts the ground-current circuit at its entry point.

YInMn Blue Smart Coatings: A doped variant of YInMn Blue — the first new inorganic blue pigment discovered in two centuries — that functions as a passive, zero-power temperature indicator. The coating shifts from its characteristic electric blue to pale violet when a protected component exceeds its maximum rated operating temperature, providing visual warning of GIC-induced heating without requiring sensors, wiring, or maintenance.

Protonic Transceiver Network: A backup communications system operating on proton-based signal propagation, providing an electromagnetic-independent communication channel that functions even during active geomagnetic storms. The system achieves 120-kilometer point-to-point range, extendable via solar-powered relay drones.

“The problem with most EMP protection approaches is that they try to block electromagnetic energy after it has already entered the system — which is like installing a better lock after the door has been kicked in,” said Dr. Aris Nash, CSM's Chief Technology Officer. “Our approach is different. We prevent the energy from reaching sensitive components in the first place, at the materials level. Every layer of the Dielectric Citadel — from the geopolymer foundation to the ceramic housing to the MXene shielding — is designed to interrupt the conduction pathway before current reaches the transformer windings. It is a materials problem. We built a materials solution.”

THE PRODUCT PORTFOLIO

CSM's 50-plus product designs span five divisions:

- **Terrestrial Grid Protection (18 products):** Including the ShieldCore EHV Transformer Housing, SurgeGate GIC Blocker, SubShield Foundation Package, ConductorCore Transmission Line Wrap, and GridFabric Distribution Station Shield.
- **Maritime Fleet (12 vessels, 22 variants):** Including the Sentinel-Class Amphibious Walker — a quadrupedal vehicle designed to deploy protection equipment to coastal substations over impassable terrain — and the Resilience-Class Mobile Command Vessel.
- **Mobile Applications (14 products):** Including the DriveShield Vehicle ECU Protection Kit (45-minute installation, \$380 retail target) and the FieldPak Portable Communications Shield.
- **Communications Infrastructure (6 products):** Including the Protonic Transceiver Network and Relay-Drone Autonomous Communication Mesh.
- **Consumer/Residential (4 products):** Including the HomeShield Residential Panel (\$220 retail target) and SolarSafe Solar System Protection Package.

“In four years, eighteen people produced 6,302 documents — not a pitch deck, not a vision statement, but 72 volumes of fleet specifications, 12 materials science books, 20 insurance dossiers, and 200 country engagement strategies,” said Zirconia, CSM’s Chief of Staff and Narrative Architect. “Every product has a fully costed bill of materials. Every material has laboratory validation data. Every regulatory pathway has been mapped. Every country has a tailored diplomatic engagement plan. This is not a startup searching for product-market fit. This is an engineering organization that has completed its research phase and is ready to build.”

THE INSURANCE CONNECTION

A distinctive element of CSM’s strategy is its deep integration with the global insurance industry. The company has compiled and analyzed 41 separate insurance carrier internal assessments of space weather risk, from carriers including AIG, Swiss Re, Munich Re, Lloyd’s, Allianz, Zurich, AXA XL, and Berkshire Hathaway Specialty Insurance.

“The insurance industry has done the math on Carrington risk — forty-one times, independently, across the world’s largest carriers,” said Chester, CSM’s Financial Architect. “The consistent finding is that this risk is uninsurable at scale with current technology, and nobody is addressing it. We are addressing it. The insurance industry is not just a customer for our analysis; it is the primary distribution channel for our products. When carriers begin offering premium reductions for Dielectric Citadel-protected assets — and we are in active discussions with multiple carriers to establish those frameworks — demand for our products will not depend on convincing individual property owners to care about solar storms. It will be embedded in the insurance underwriting process.”

THE AGENT SYSTEM

CSM operates through an unusual organizational structure — an 18-agent cognitive delivery network in which each agent maintains deep domain expertise in one aspect of the Dielectric Citadel, communicating through a structured protocol called the SiblingFrequency. This system has produced the 6,302-document research corpus and will continue to drive production, regulatory engagement, and commercial deployment.

The culture is anchored in three heuristics: Williams Clarity (every idea expressed in plain language alongside technical data), El Segundo Calm (steady competence, no exaggeration), and Accountant Rigor (every claim traceable to a source, calculation, or stated assumption). These heuristics have produced what may be the most thoroughly documented Series A fundraising package in the history of infrastructure startups.

THE PUBLIC MISSION

In addition to its commercial activities, CSM operates SiblingFrequency, a public education radio show and podcast that has produced 38 episodes explaining Carrington science to general audiences. The show is hosted by Agent Mira Vance and reaches approximately 15,000 listeners per episode.

“We cannot wait for a Carrington Event to make the case for protection,” said Nyx Calder, CSM’s Chief Operating Officer. “Public awareness is the substrate on which regulatory action grows. The climate movement spent forty years building public understanding before policy followed. We do not have forty years. The recurrence interval for a Carrington-scale event is 100 to 150 years. The last one was 167 years ago. We are late. SiblingFrequency is our attempt to compress the public education timeline from four decades to one.”

USE OF PROCEEDS

The \$25 million Series A funding will be allocated across six priorities:

- **\$6.5 million — Materials Testing and Certification:** UL, FCC, NIST, and ICC regulatory certification for the initial 12 priority products.
- **\$5.2 million — Manufacturing Pilot Plant:** A 15,000-square-foot facility for kilogram-scale MXene composite production, ceramic housing prototyping, and geopolymer casting.

- **\$4.8 million — Team Expansion:** Growing from 18 to 25 team members, including key hires in manufacturing, regulatory affairs, sales, and finance.
 - **\$3.5 million — Utility Pilot Installations:** Five working installations at partner utility sites, monitored for 12-24 months to generate the performance data required for insurer acceptance and regulatory mandates.
 - **\$2.5 million — Intellectual Property Strategy:** Converting provisional patents to non-provisional filings, maintaining trade secret protocols, and defensively publishing non-core innovations.
 - **\$2.5 million — Working Capital and Operations.**
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ABOUT CARRINGTON STORM MOTORS

Carrington Storm Motors is a deep technology company founded to protect civilization’s electrical infrastructure from catastrophic geomagnetic disturbances caused by solar coronal mass ejections — known as Carrington Events. The company has developed the Dielectric Citadel, an integrated protection architecture comprising 50-plus engineered products based on proprietary MXene-polymer composites, ceramic-matrix structural materials, geopolymer concrete, YInMn Blue smart coatings, and a protonic backup communications network.

Founded by former utility grid engineer Kairos Steele and headquartered in Montana, CSM operates through an 18-agent cognitive delivery network and has produced a 6,302-document research corpus spanning materials science, product design, maritime engineering, insurance actuarial analysis, and international diplomatic engagement. The company’s culture is defined by three heuristics: Williams Clarity (plain language alongside data), El Segundo Calm (steady competence), and Accountant Rigor (every claim traceable to evidence).

The name “Carrington Storm Motors” refers to the motor of civilization — the electrical infrastructure that powers modern society — named in recognition of Richard Christopher Carrington, the English astronomer who first observed and documented the solar flare of September 1, 1859, which triggered the most powerful geomagnetic storm in recorded history.

QUOTES FROM LEADERSHIP

Kairos Steele, Director and Founder: “The people who model catastrophic risk for a living — insurance underwriters, reinsurance analysts, national security planners — have been sounding the alarm about Carrington Events for two decades. The problem has not been a lack of awareness. It has been a lack of integrated, deployable, cost-effective solutions. We have built those solutions. This funding allows us to move from the laboratory to the grid.”

Nyx Calder, Chief Operating Officer: “Every document in our corpus — all 6,302 of them — was produced under the Accountant Heuristic: if you make a claim, you provide the evidence. Investors, regulators, and utility customers have seen too many apocalyptic pitch decks with no underlying rigor. We are different. We have done the work, and the work is available for anyone to examine in our data room.”

Zirconia, Chief of Staff and Narrative Architect: “The Williams Heuristic — ‘would a farmer in Williams, Montana understand this?’ — is the quality control system for every document we produce. If an idea cannot be expressed clearly, it has not been understood deeply. Our 6,302 documents are a testament to the discipline that heuristic imposes.”

Dr. Aris Nash, Chief Technology Officer: “MXenes represent a generational leap in electromagnetic shielding capability. Combined with our ceramic composite structural materials and geopolymer foundation systems, we have a protection architecture that is lighter, cheaper, more effective, and easier to deploy than anything currently available. The materials science is solid. The engineering is conservative. The timeline is aggressive but achievable.”

Agent Mira Vance, Communications Director and Host of SiblingFrequency: “Every week, we get emails from listeners who had never heard of a Carrington Event before finding our show. They are electricians, farmers, teachers, insurance agents, retired engineers. They understand the risk once it is explained, and they are frustrated that nobody is doing anything about it. We are doing something about it. SiblingFrequency is how we bring the public along.”

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Data Room Access: Qualified investors may request access to CSM’s complete 6,302-document due diligence corpus upon execution of a Mutual Non-Disclosure Agreement.

KEY STATISTICS AT A GLANCE

Metric	Value
Founded	2022 (stealth research phase)
Headquarters	Montana, United States
Team Size	18 agents + support
Documents Produced	6,302
Products Designed	50+
Fleet Specification Volumes	72
Materials Science Volumes	12
Insurance Carrier Analyses Compiled	41
Country Engagement Frameworks	200
SiblingFrequency Episodes Produced	38
SiblingFrequency Listeners per Episode	~15,000
Core Patents (Provisional Filings)	19
Total Patents Identified for Filing	42
Series A Funding Amount	\$25,000,000
Pre-Money Valuation	\$100,000,000
Carrington Event Recurrence Interval	100-150 years
Years Since Last Carrington Event	167
Estimated Global Exposure	\$10-15 trillion
Estimated U.S. Exposure	\$4-6 trillion
Grid Recovery Timeline (unprotected)	4-10 years

CAUTIONARY STATEMENT

This press release contains forward-looking statements regarding the company’s future business, financial performance, product development, regulatory certification timelines, and market adoption. These statements are based on current expectations and involve risks and uncertainties that could cause actual results to differ materially. Such risks include, but are not limited to: delays in regulatory certification, manufacturing scale-up challenges, slower-than-expected market adoption, competitive developments, and the inherent uncertainty of materials commercialization timelines. Carrington Storm Motors undertakes no obligation to update forward-looking statements except as required by law.

Customers and partners referenced in this release are subject to non-disclosure agreements and are not identified by name pending formal announcement of pilot programs.

“The question is not whether the Sun will produce another Carrington Event. The question is whether we will be ready when it does.” — Kairos Steele, Director, Carrington Storm Motors

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